

# Dongho Lee

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## Research Interests

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**Keywords :** Symplectic Geometry and Dynamics, Three-body Problem

My primary research focuses on understanding dynamical systems through the framework of symplectic geometry. I am particularly interested in classical Hamiltonian systems such as the three-body problem, where families of periodic orbits exhibit rich bifurcation phenomena. Using tools from Floer theory, I investigate the geometric and topological mechanisms underlying these behaviors, and I am also interested in connecting these theoretical insights to practical applications in celestial mechanics and related areas.

## Education

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### Ph.D. in Mathematics

Department of Mathematical Science, Seoul National University (SNU)    Mar 2018 – Feb 2025

### Bachelor of Science in Mathematics

### Bachelor of Economics in Economics, Minor in Physics

College of Liberal Studies, SNU

Mar 2013 – Feb 2018

Graduated with Honors

## Research Experience

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### Postdoctoral Researcher

Center for Quantum Structures in Modules and Spaces (QSMS), SNU    May 2025 – Present

Research Institute of Mathematics, SNU

Mar 2025 – Apr 2025

Postdoctoral Supervisor: Prof. Cheol-hyun Cho

### Graduate Student Researcher

Department of Mathematical Sciences, SNU

Mar 2018 – Feb 2025

Ph.D Advisor: Prof. Otto van Koert

## Publications

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### 1. Conley-Zehnder Indices of Spatial Rotating Kepler Problem

arXiv 2506.14325

*Journal of Topology and Analysis*, 27 Feb 2026

### 2. Global Hypersurfaces of Section and the Spatial Kepler Problem

Advisor: Prof. Otto van Koert

Ph.D. Thesis, Dec 2024

### 3. Global Hypersurfaces of Section for Geodesic Flows on Convex Hypersurfaces

arXiv 2401.20667, with Sunghae Cho

*Archiv der Mathematik*, Vol. 123, 31 Jul 2024

## Talks and Presentations

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Sildes and posters are available at my homepage.

### Invited Talks

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1. **Bifurcation of Periodic Orbits in Hamiltonian Systems**  
G-Lamp Mathematics Seminar, Soongsil University, Seoul, Korea 22 Jan 2026
2. **Generic Bifurcation of Hamiltonian Orbits**  
Symplectic Learning Seminar, SNU, Seoul, Korea 12 Dec 2025
3. **Hamiltonian Dynamics and Three-body Problem**  
Rookie's Pitch, SNU, Seoul, Korea 14 Oct 2025
4. **Three-body Problem and Related Problems**  
QSMS 2025 Summer Workshop, Goseong, Korea 21 Aug 2025
5. **Periodic Orbits of Rotating Kepler Problem**  
Billiards and Quantitative Symplectic Geometry, Heidelberg, Germany 16 Jul 2025
6. **Closed orbits of the spatial rotating Kepler problem**  
2025 Korean Mathematical Society Spring Meeting, Daejeon, Korea 25 Apr 2025

### Poster Talks

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1. **Periodic Orbits and Symplectic Invariants in the Rotating Kepler Problem**  
Celestial Mechanics & N-body Dynamics, Tokyo, Japan 16 Nov 2025

### Conference Participation

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|--------------------------------------------------------------------------------------------------------------|---------------------|
| <b>International Conference on Celestial Mechanics</b><br>SUSTech, ShenZhen, China                           | 20 – 24 Apr 2026    |
| <b>New Developments in Symplectic Geometry</b><br>Seoul National University, Seoul, Korea                    | 25 – 28 Nov 2025    |
| <b>East Asian Symplectic Conference 2025</b><br>Hokkaido University, Sapporo, Japan                          | 25 – 29 Sep 2025    |
| <b>2025 Symplectic Retreat</b><br>Stanford Hotel, Tongyeong, Korea                                           | 23 – 26 Jan 2025    |
| <b>From Hamiltonian Dynamics to Symplectic Topology and Beyond</b><br>Institut Henri Poincaré, Paris, France | 3 – 7 Jun 2024      |
| <b>Topology of 4-manifolds and Related Topics</b><br>Ocean Suites Hotel, Jeju, Korea                         | 21 – 27 Jan 2024    |
| <b>East Asian Symplectic Conference 2023</b><br>Ocean Suites Hotel, Jeju, Korea                              | 29 Oct – 4 Nov 2023 |
| <b>QSMS Workshop on Symplectic Geometry and Related Topics</b><br>Ocean Suites Hotel, Jeju, Korea            | 5 – 10 Feb 2023     |
| <b>Workshop on Symplectic Dynamics</b><br>Instituto Superior Técnico, Lisboa, Portugal                       | 27 Jun – 1 Jul 2022 |

**Symplectic Geometry and Beyond (Part I)**

7 – 10 Feb 2022

Ocean Suite Hotel, Jeju, Korea

**Fukaya 60 Geometry and Everything**

17 – 22 Feb 2019

Kyoto University, Kyoto, Japan

**Personal Affiliations****Korean Mathematical Society, Korea**

2025 – Present

**QSMS-BK21 Symplectic Seminar, Korea**

2021 – Present

**Teaching Experience****Teaching and Course Assistant, SNU**

Mar 2018 – Feb 2025

**Major Courses****Algebraic Topology (Graduate Course)**

Fall 2019

**Introduction to Topology**

Spring 2023, Spring 2020

**Introduction to Differential Geometry**

Spring, Fall 2021

**Selected Topics Seminar (College of Liberal Studies)**

Spring 2023

**General Education Courses****Head Teaching Assistant for Introductory Mathematics Courses**

Spring 2020

**Engineering Mathematics**

Fall 2024, Fall, Summer, Spring 2023, Fall 2020

**Calculus for Life Science**

Fall 2022, Spring 2021

**Fundamentals and Applications of Mathematics**

Spring 2022

**Calculus Practice**

Fall, Spring 2018

**Calculus**

Fall, Spring 2018

**Awards and Honors****Merit-based Scholarship, Department of Mathematical Sciences, SNU**

2018

**National Scholarship, Korean Student Aid Foundation**

2015

**Merit-based Scholarship, College of Liberal Studies, SNU**

2013

**Languages****Korean:** Native**English:** Fluent**Japanese, French:** Basic – Intermediate**References**

Prof. Otto van Koert

Department of Mathematical Sciences, SNU

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Relationship: Ph.D. Advisor

Prof. Cheol-hyun Cho

Department of Mathematics, POSTECH

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Relationship: Postdoctoral Supervisor